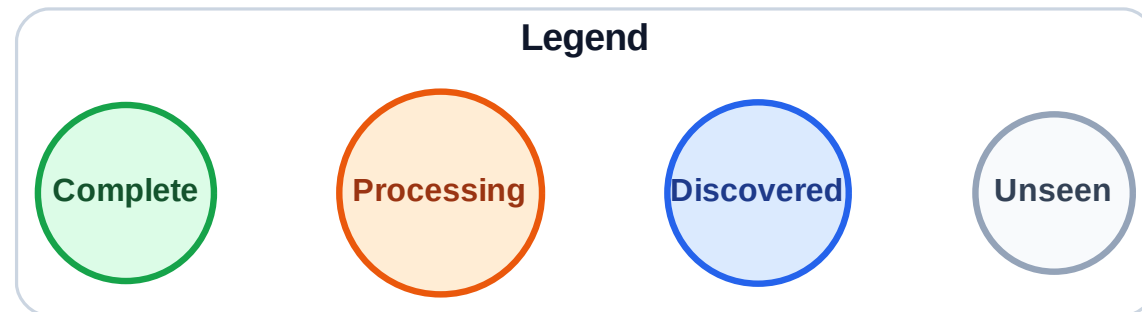
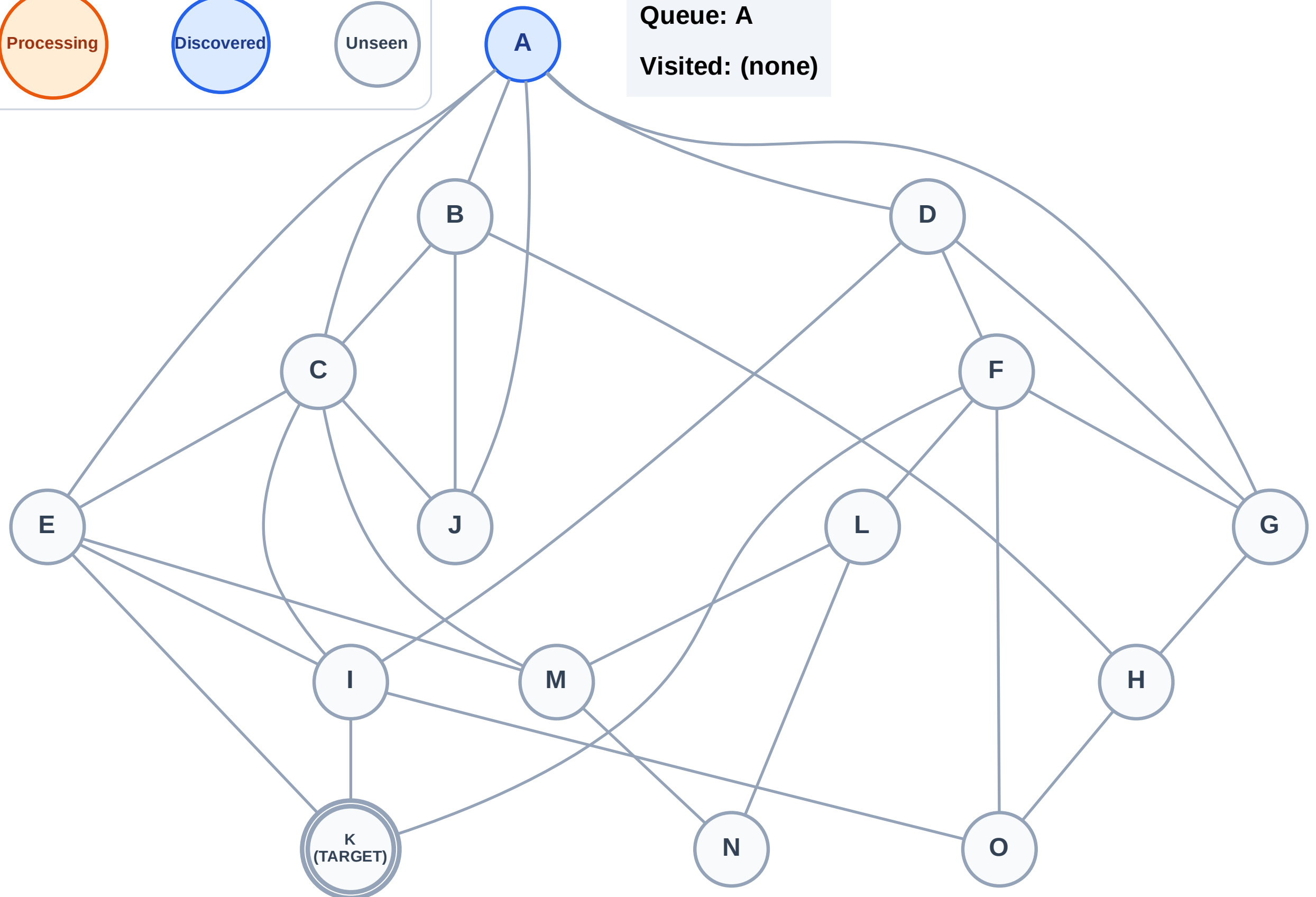


BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

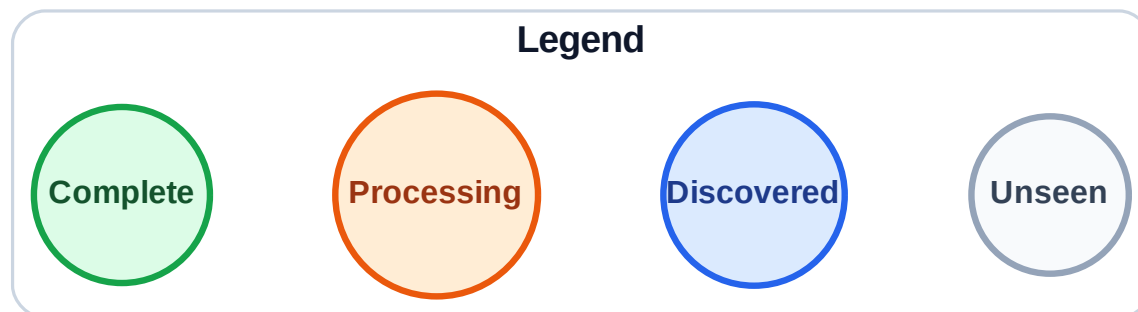


Queue: A
Visited: (none)

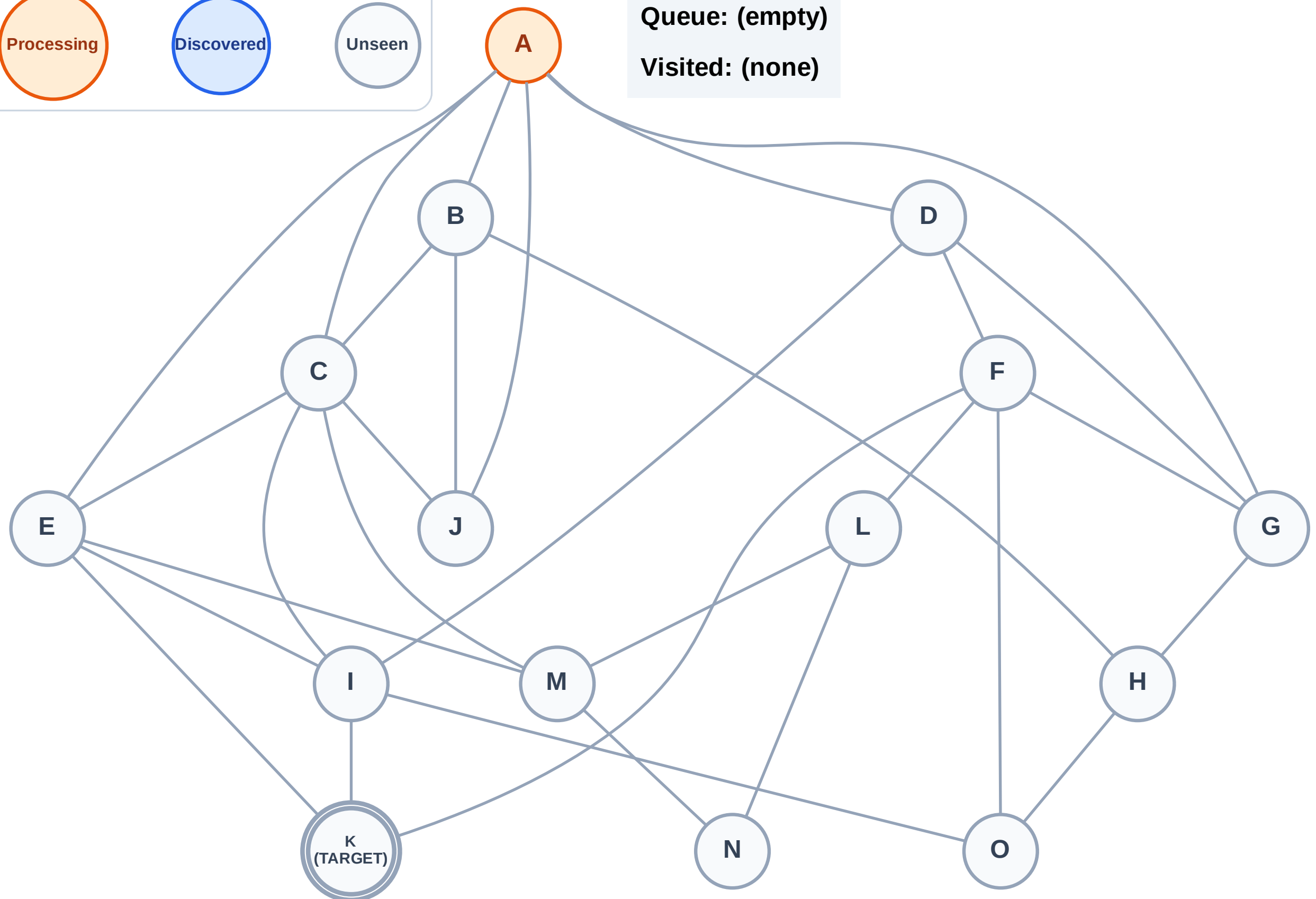


BFS Traversal

Step 2: Process node A

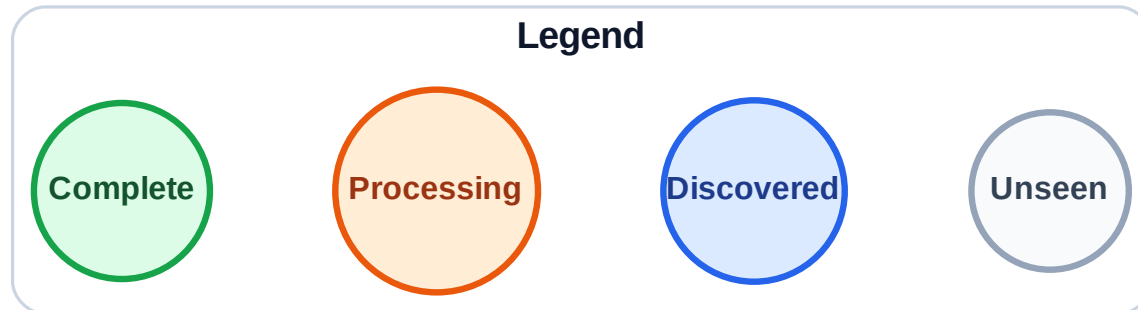


Queue: (empty)
Visited: (none)

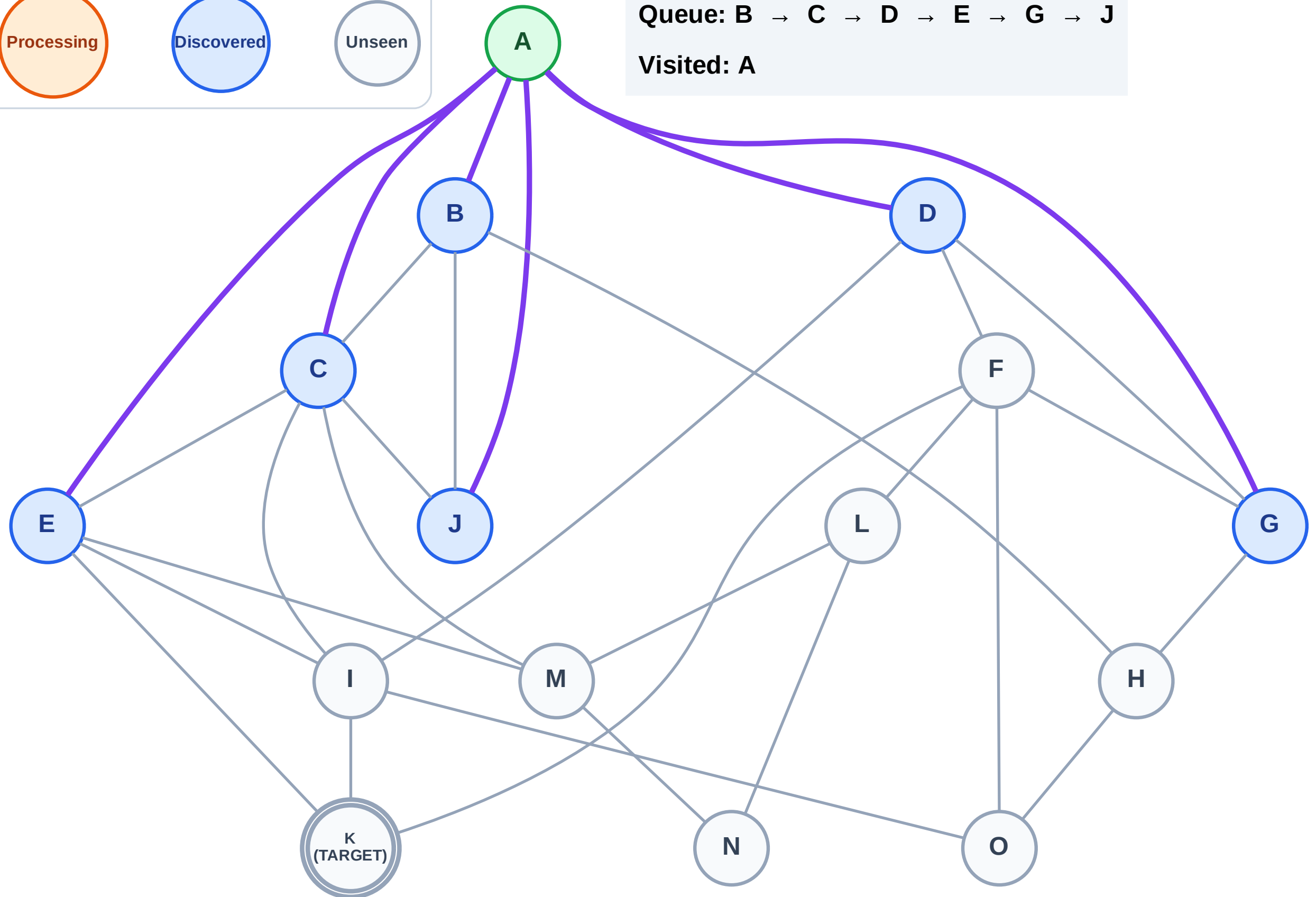


BFS Traversal

Step 3: Discover B, C, D, E, G, J from A

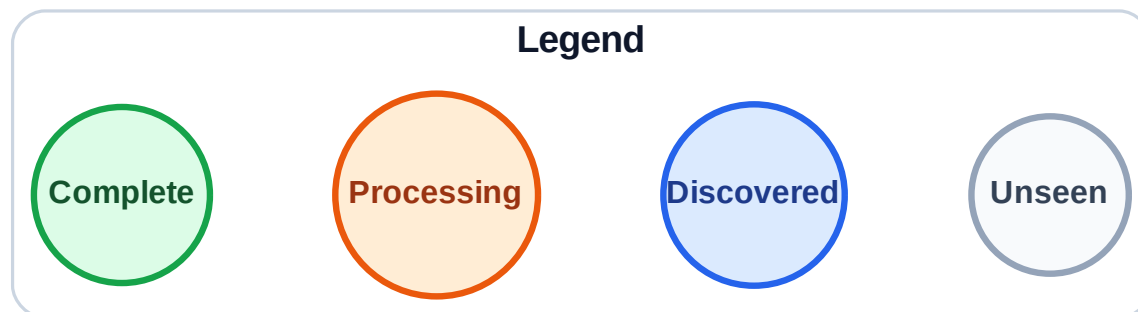


Queue: B → C → D → E → G → J
Visited: A

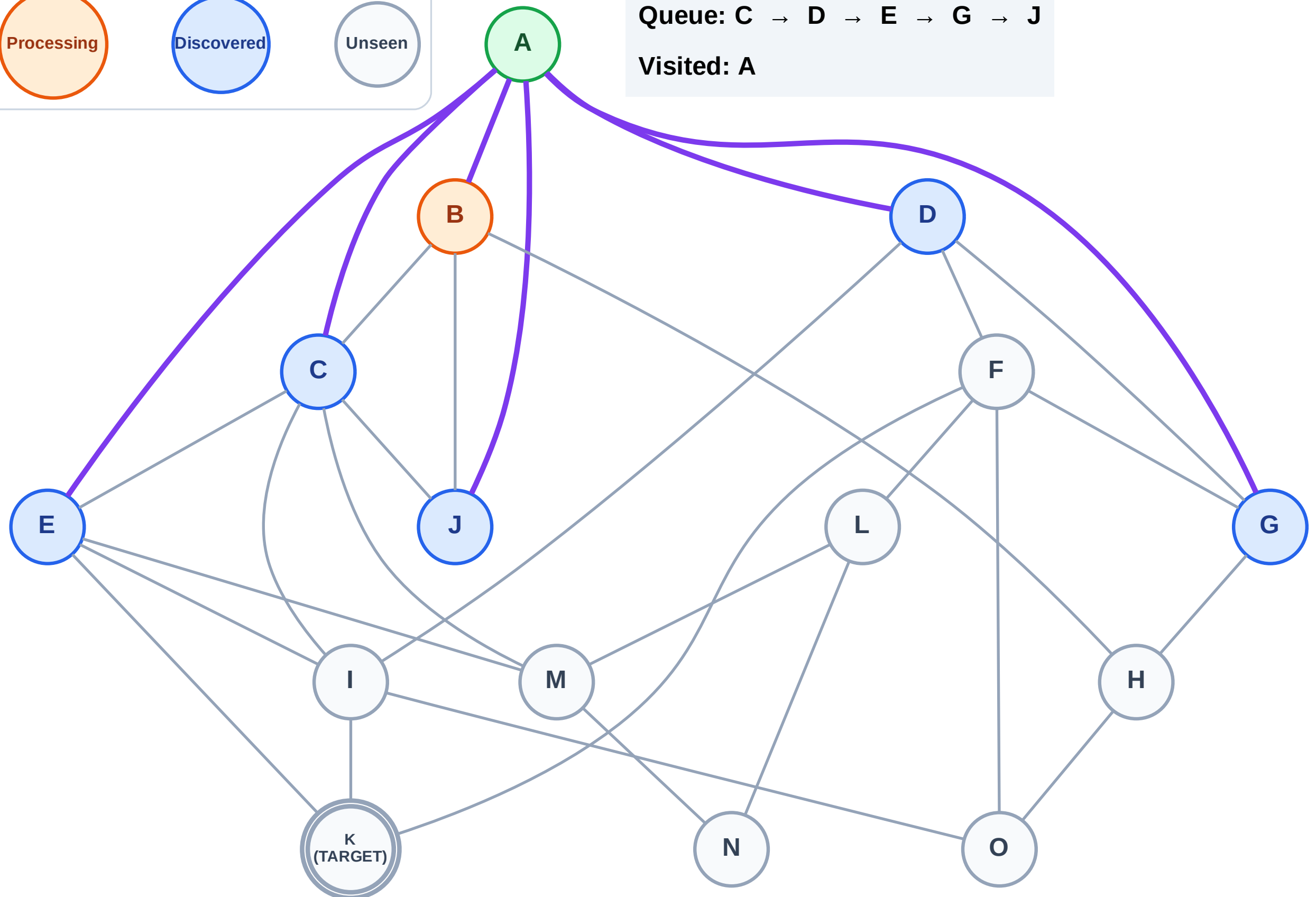


BFS Traversal

Step 4: Process node B



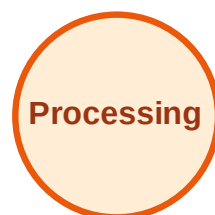
Queue: C → D → E → G → J
Visited: A



BFS Traversal

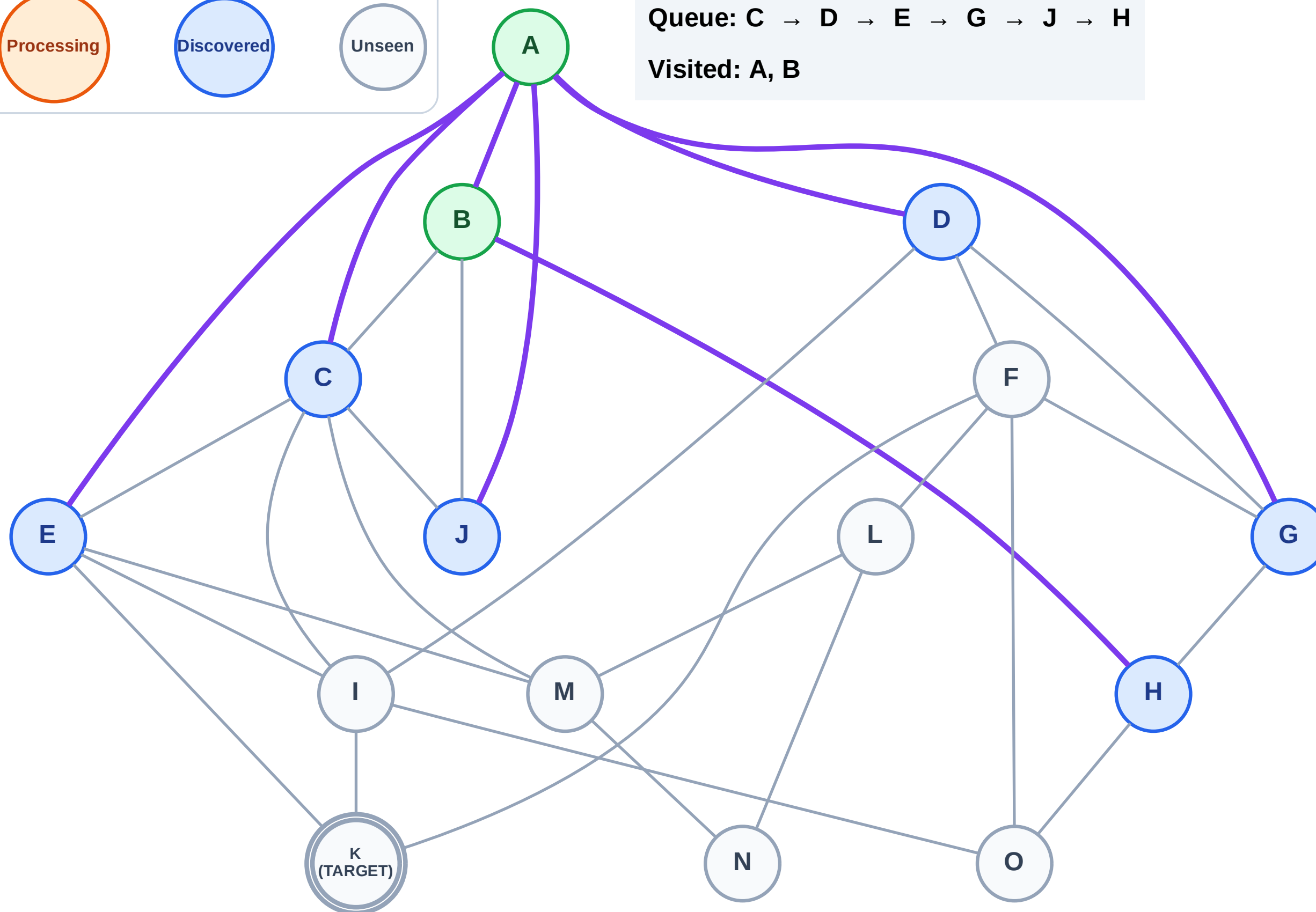
Step 5: Discover H from B

Legend



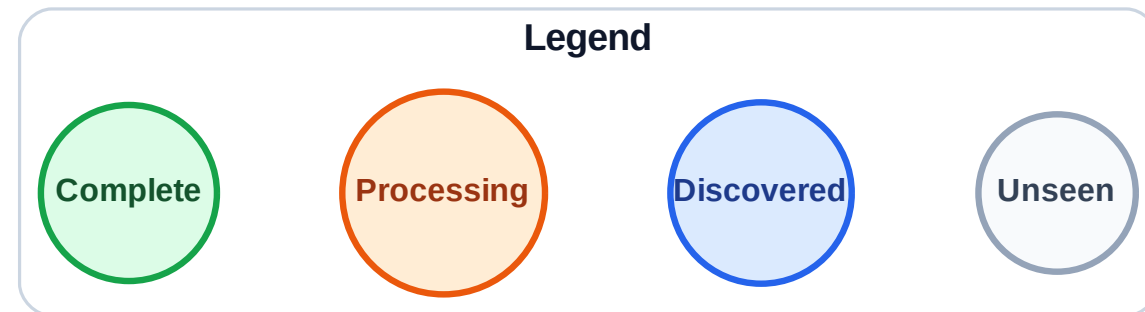
Queue: C → D → E → G → J → H

Visited: A, B



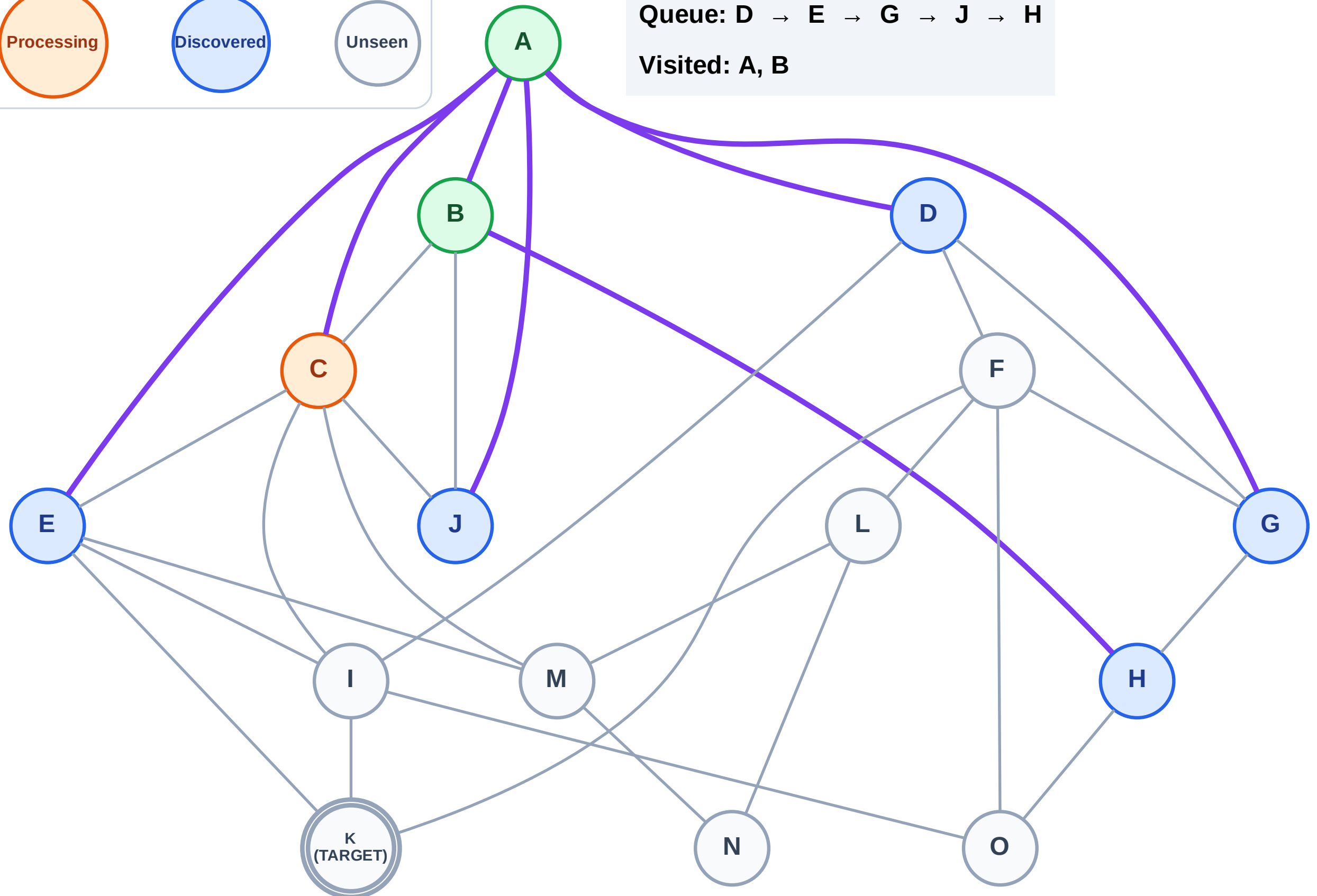
BFS Traversal

Step 6: Process node C



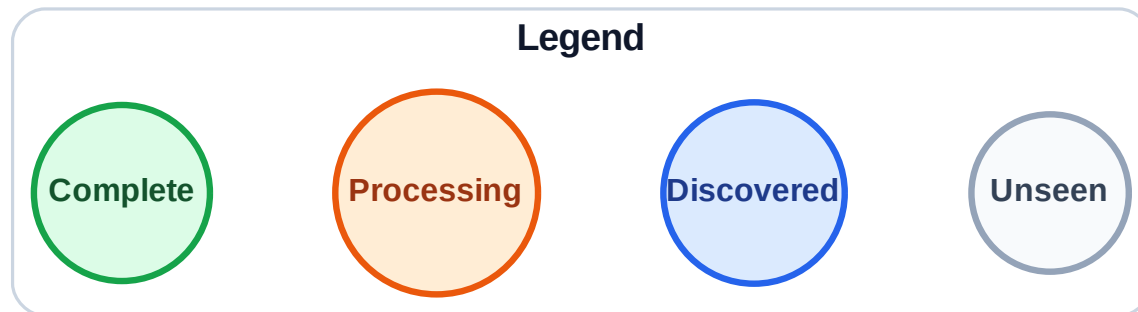
Queue: D → E → G → J → H

Visited: A, B



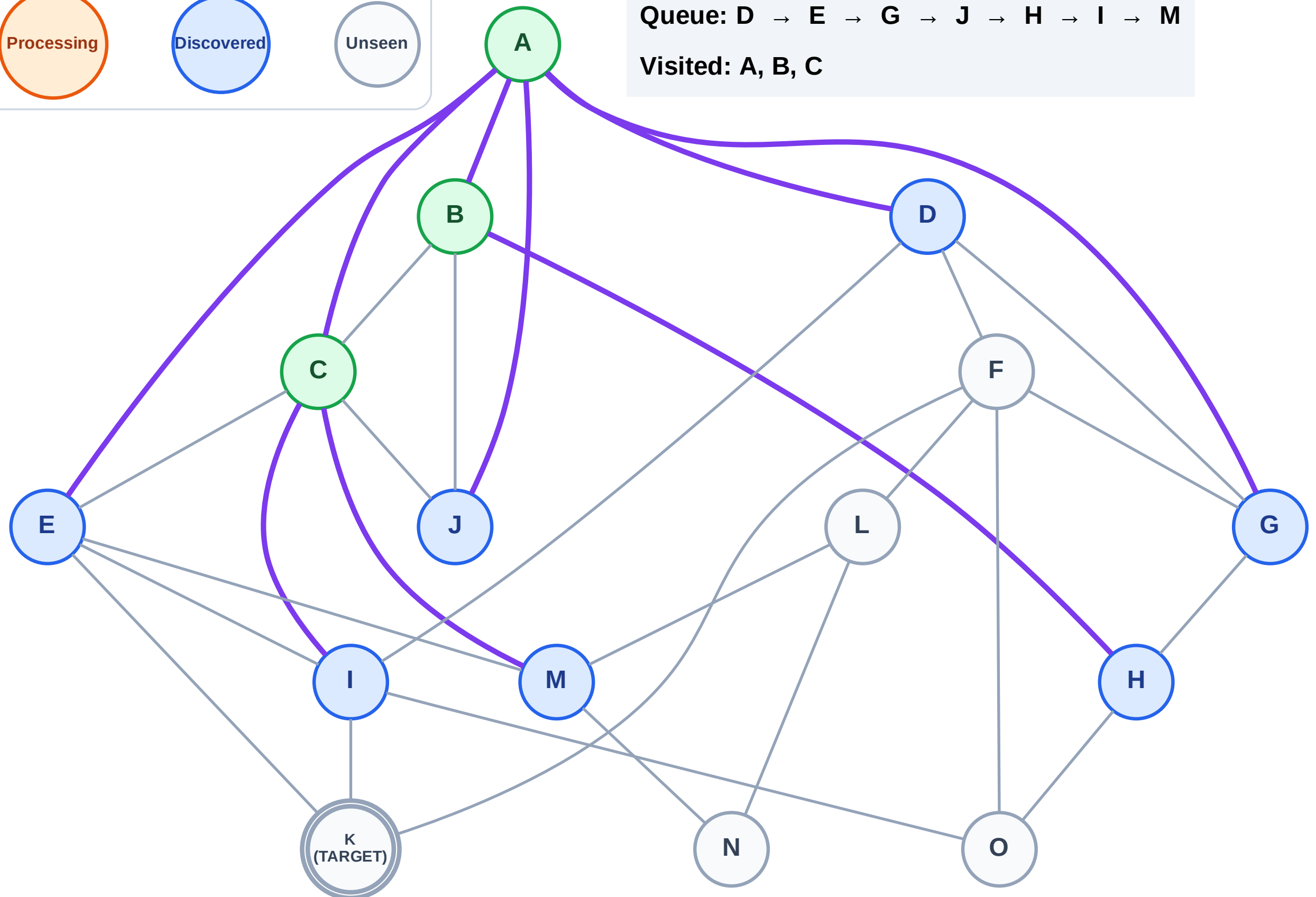
BFS Traversal

Step 7: Discover I, M from C



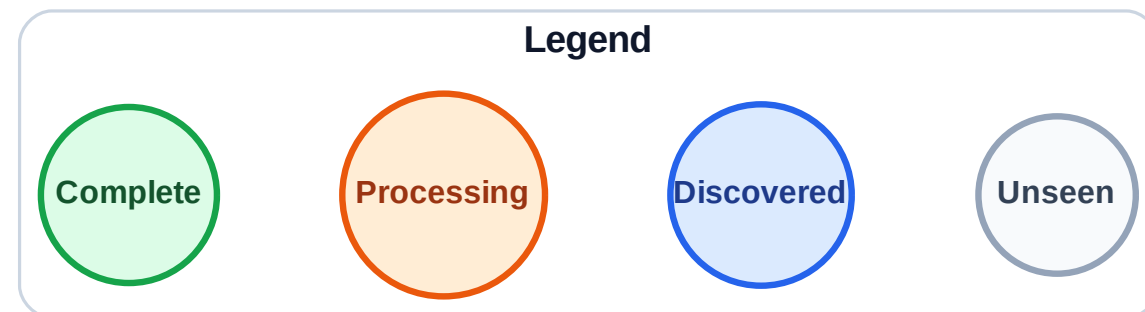
Queue: D → E → G → J → H → I → M

Visited: A, B, C



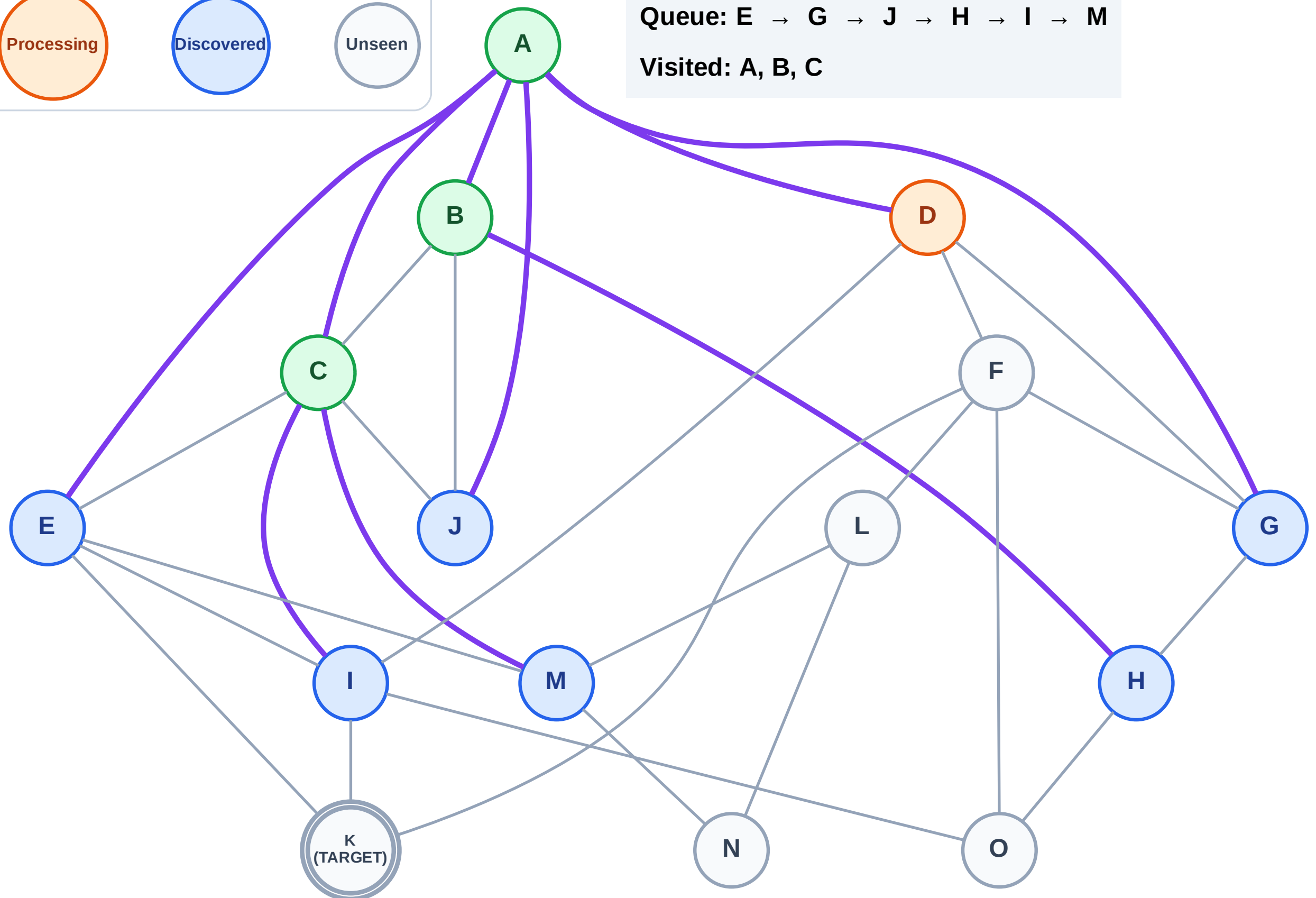
BFS Traversal

Step 8: Process node D



Queue: E → G → J → H → I → M

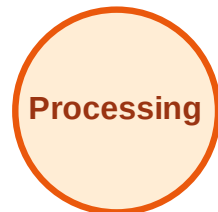
Visited: A, B, C



BFS Traversal

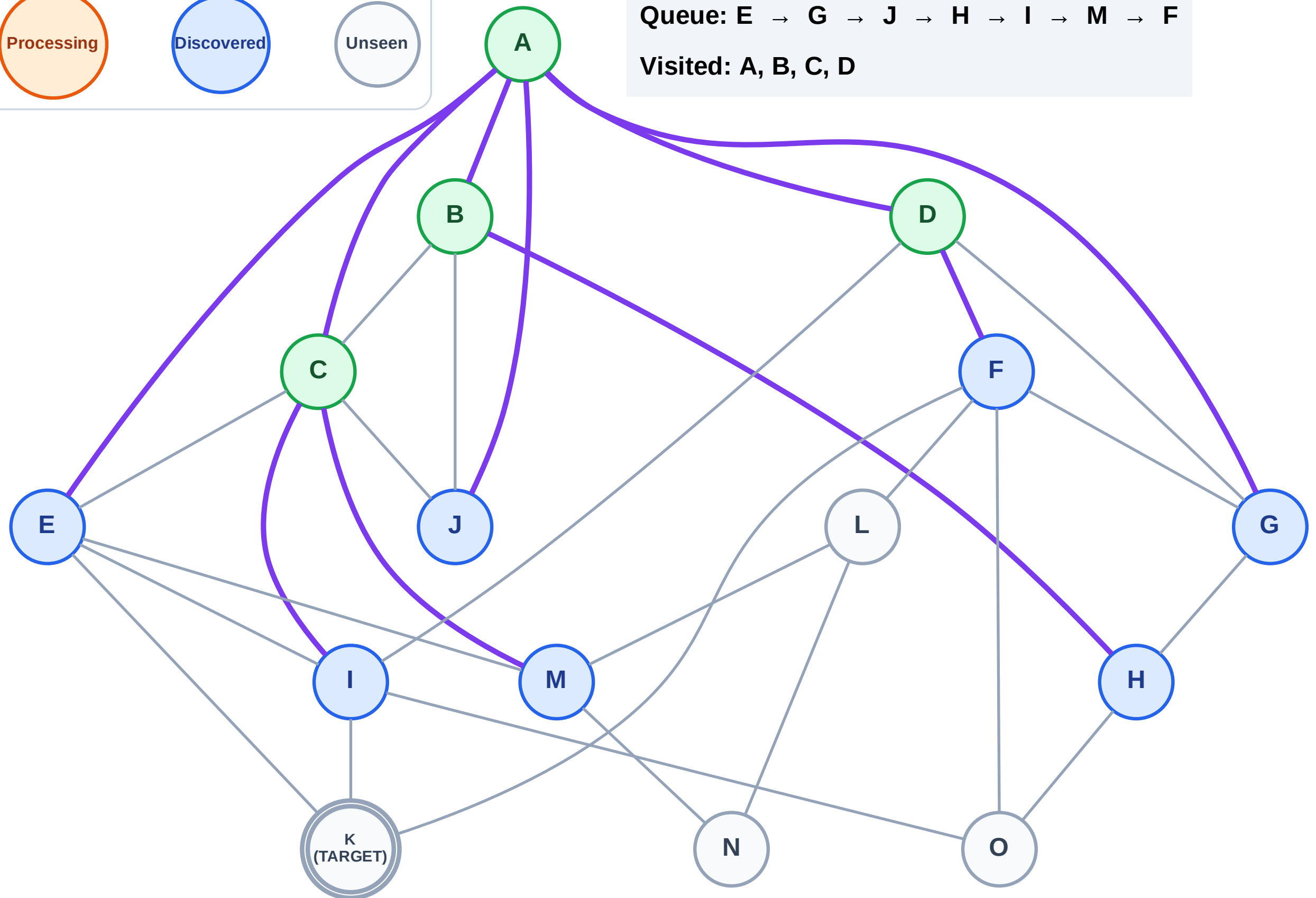
Step 9: Discover F from D

Legend



Queue: E → G → J → H → I → M → F

Visited: A, B, C, D



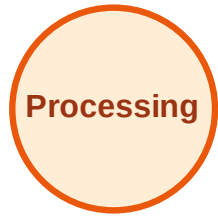
BFS Traversal

Step 10: Process node E

Legend



Complete



Processing



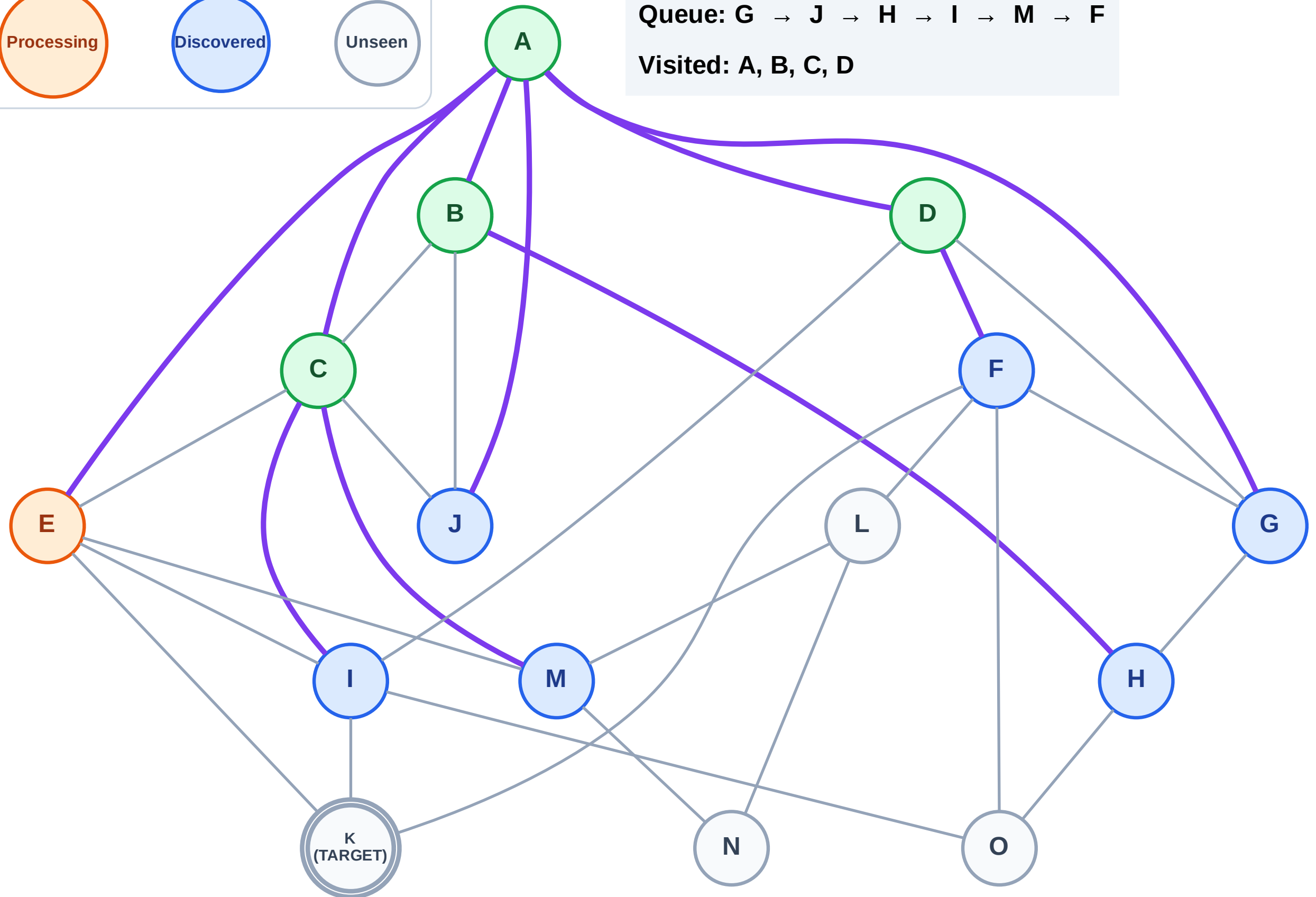
Discovered



Unseen

Queue: G → J → H → I → M → F

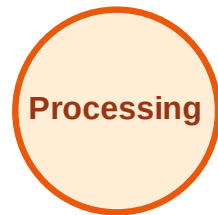
Visited: A, B, C, D



BFS Traversal

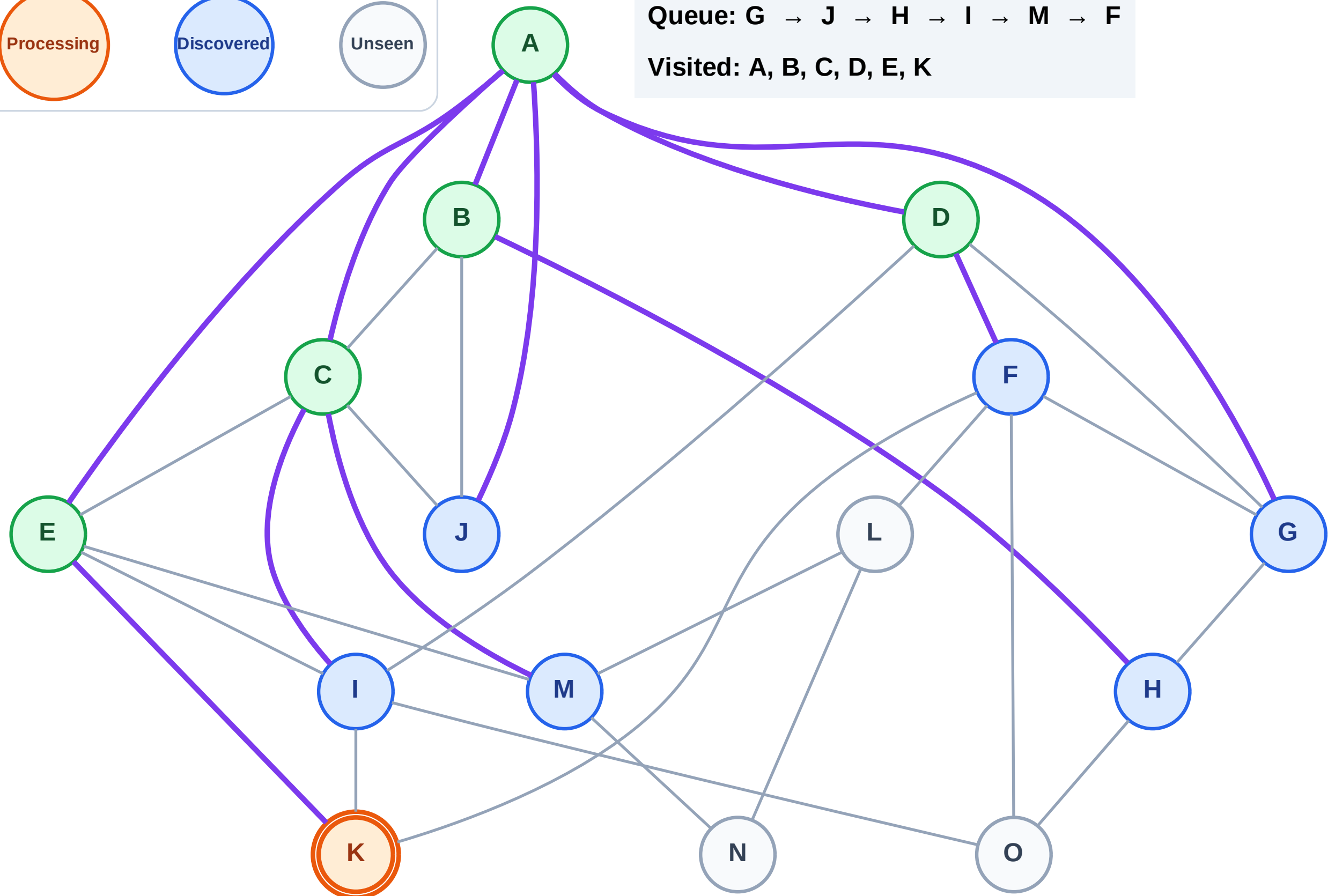
Step 11: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → E → K

Legend



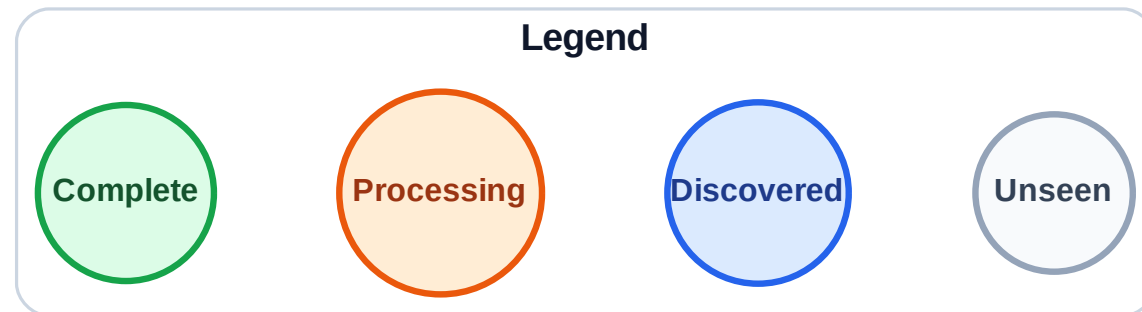
Queue: G → J → H → I → M → F

Visited: A, B, C, D, E, K

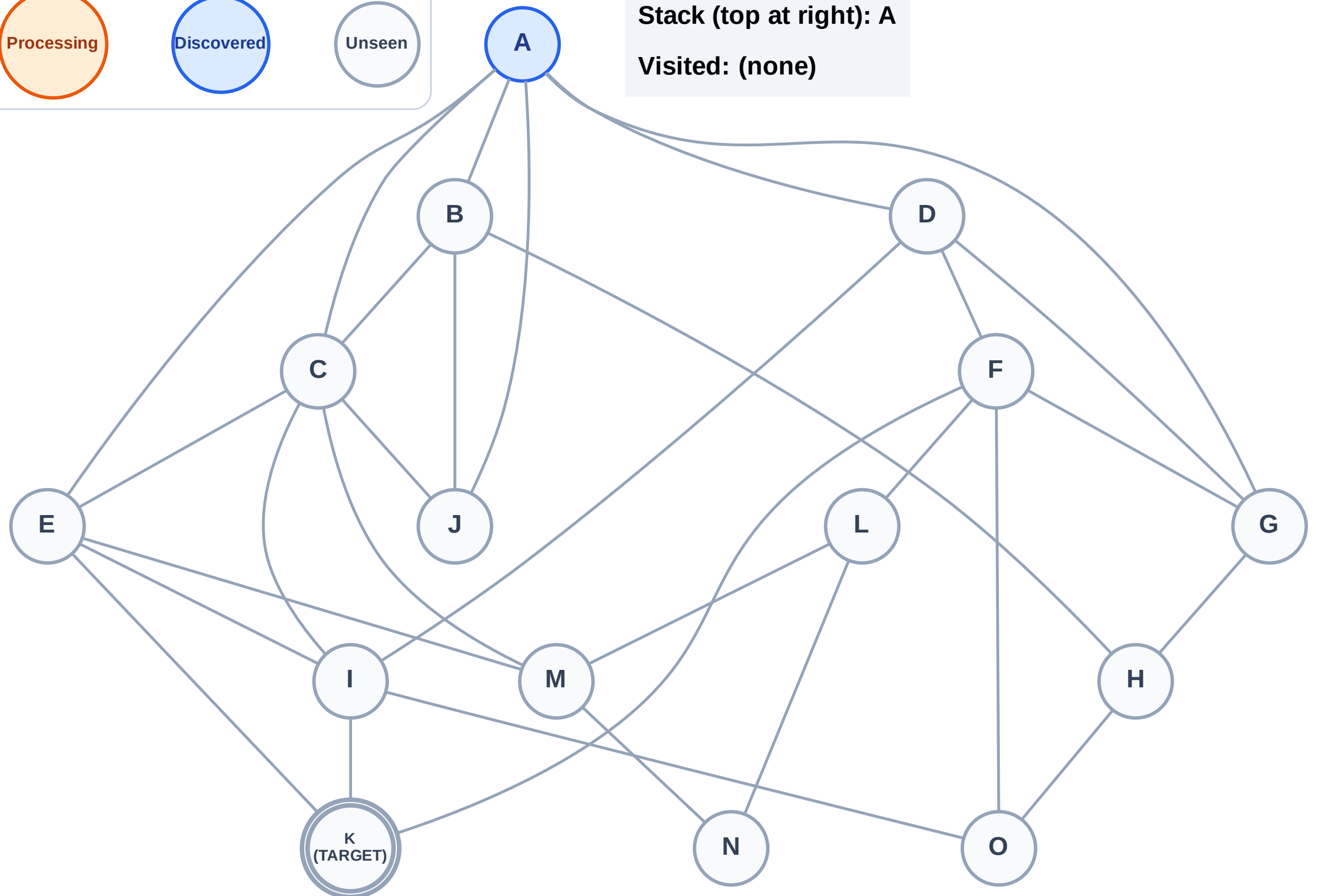


DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))



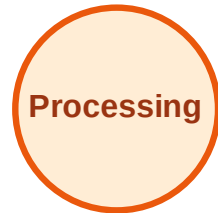
Stack (top at right): A
Visited: (none)



DFS Traversal

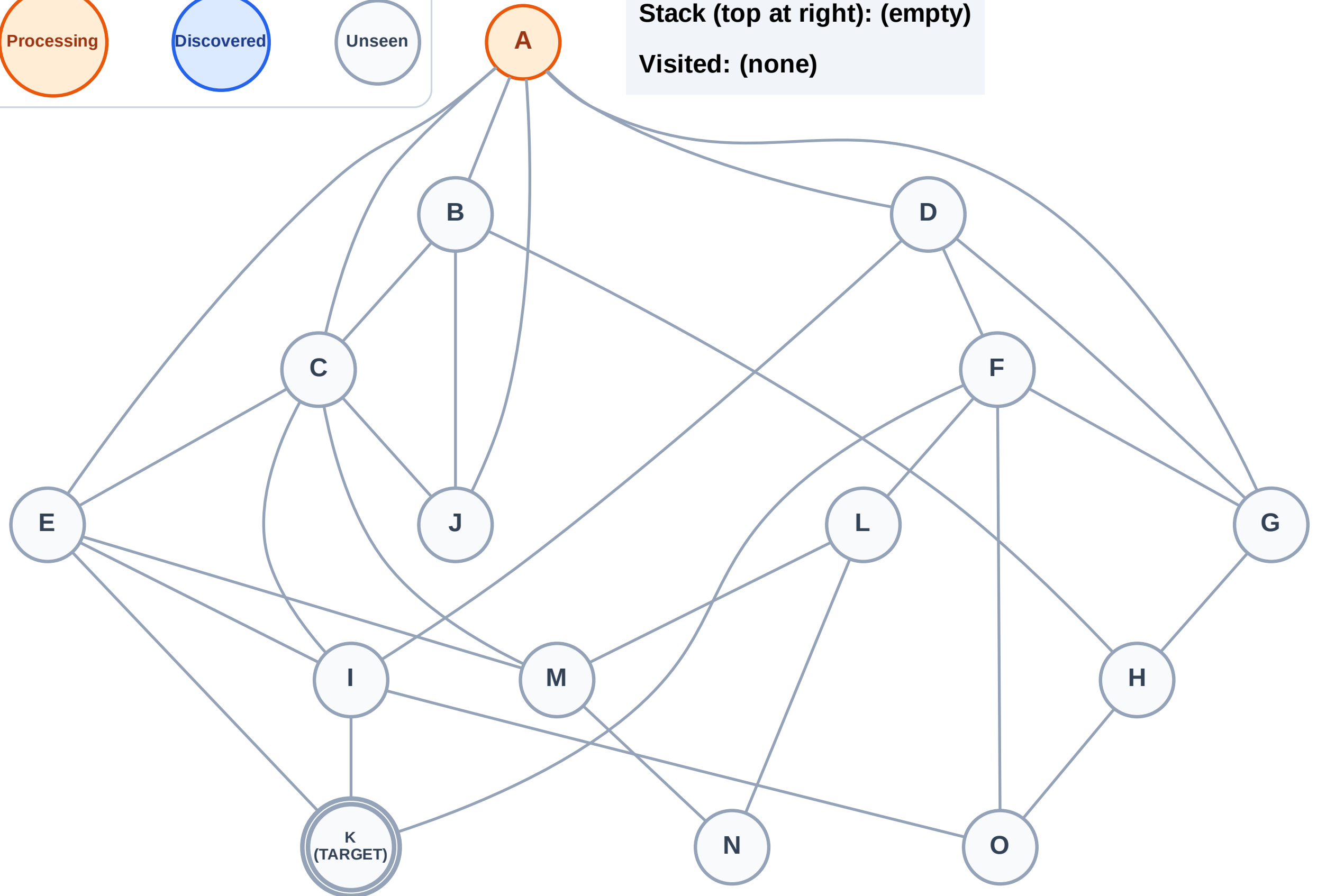
Step 2: Process node A

Legend



Stack (top at right): (empty)

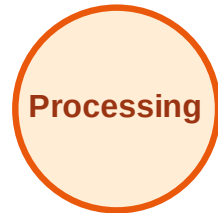
Visited: (none)



DFS Traversal

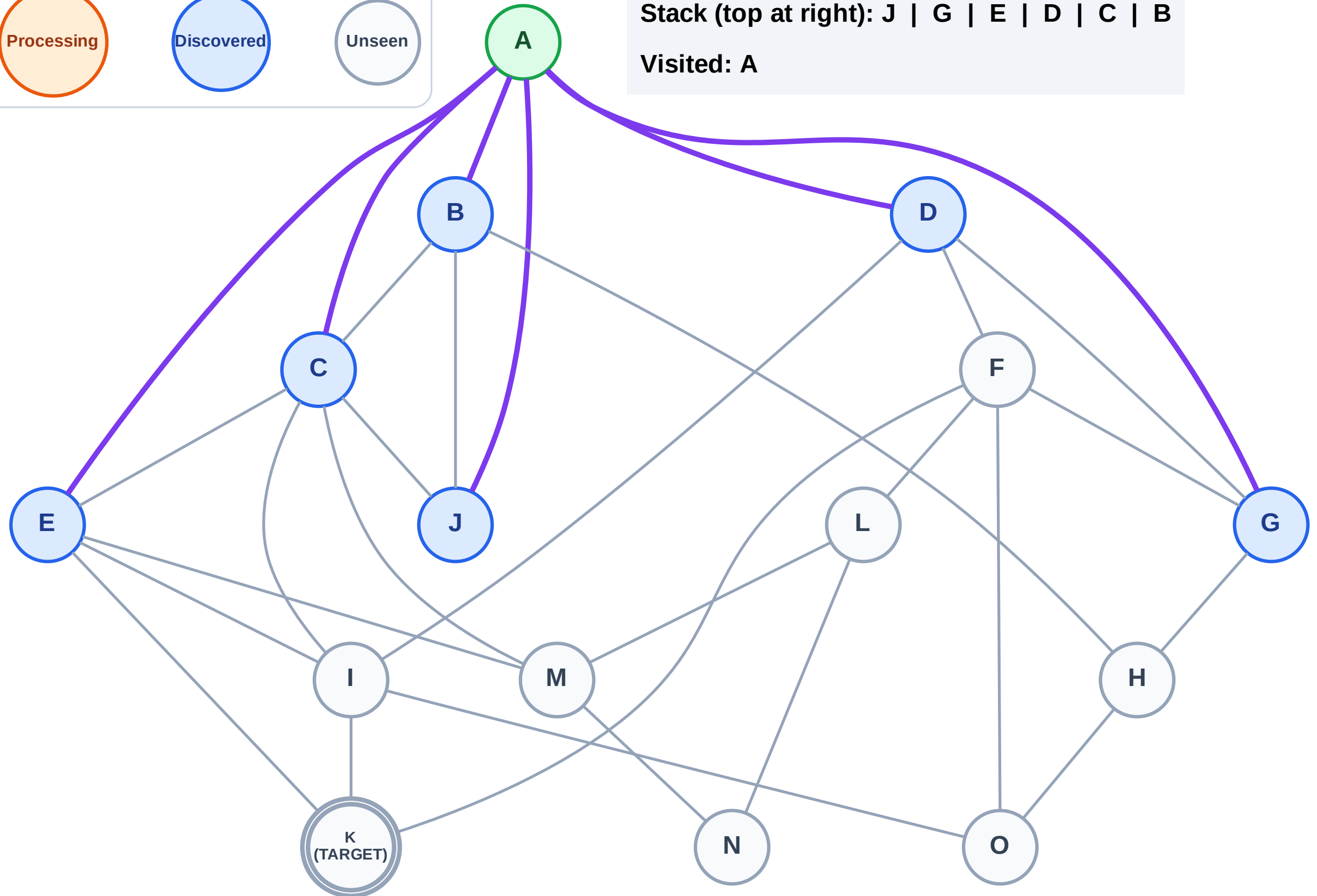
Step 3: Discover J, G, E, D, C, B from A

Legend



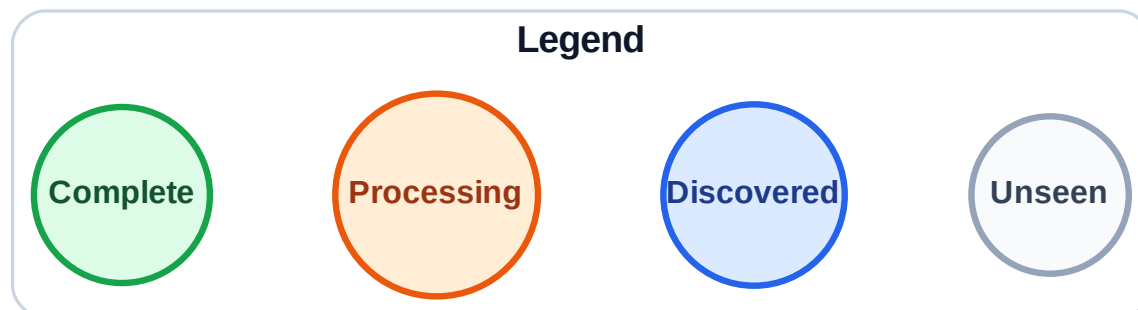
Stack (top at right): J | G | E | D | C | B

Visited: A

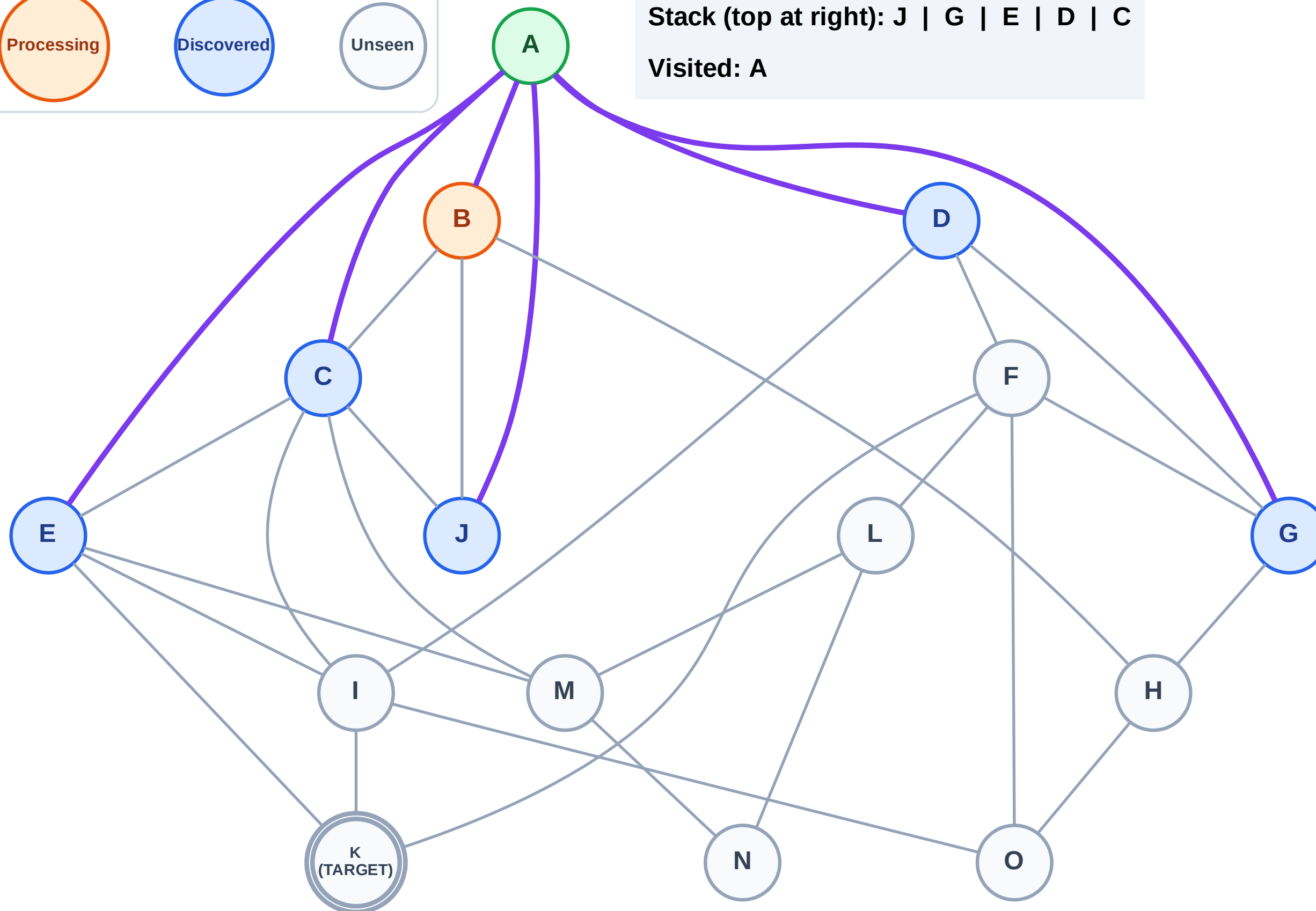


DFS Traversal

Step 4: Process node B



Stack (top at right): J | G | E | D | C
Visited: A



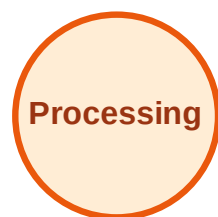
DFS Traversal

Step 5: Discover H from B

Legend



Complete



Processing



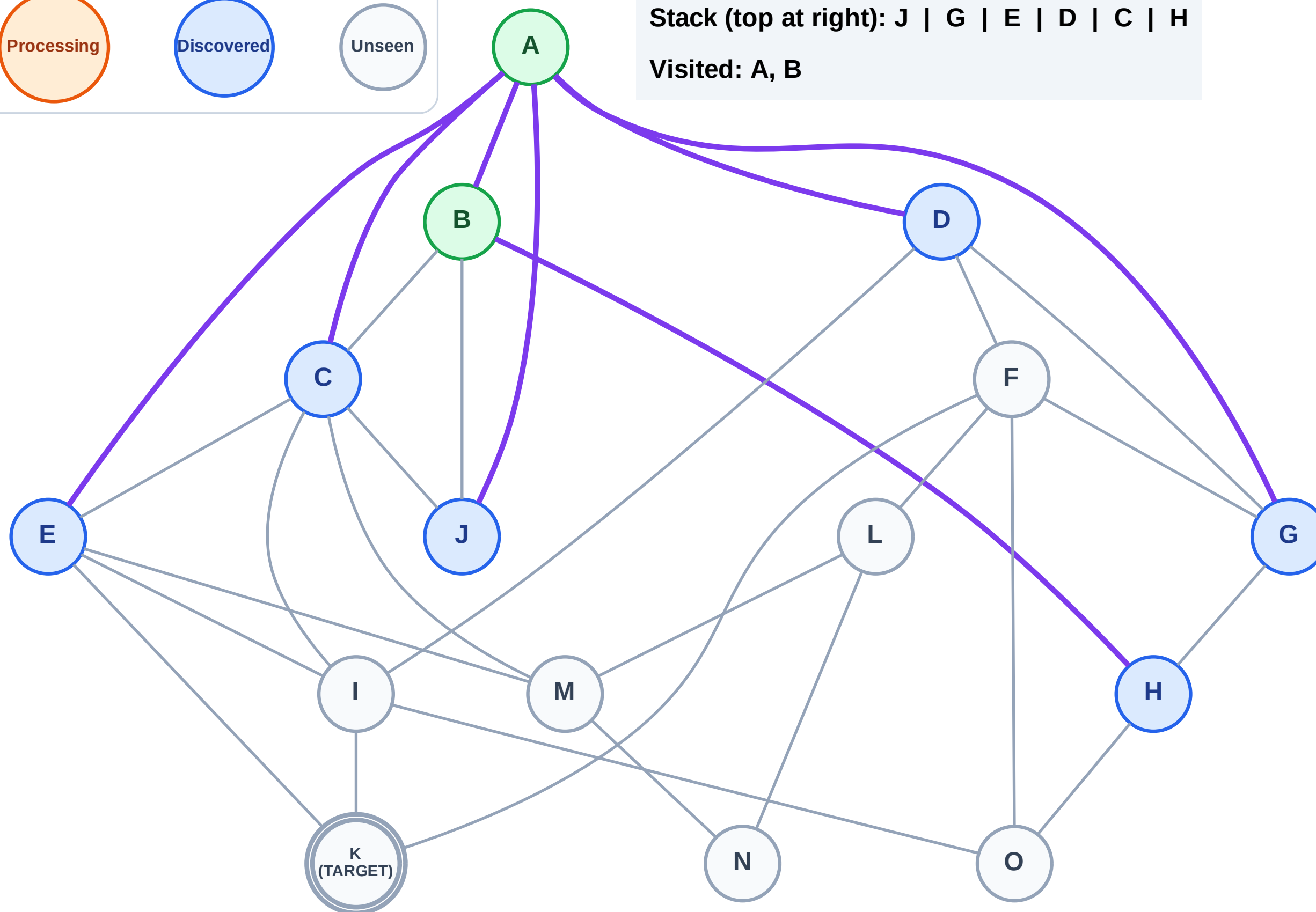
Discovered



Unseen

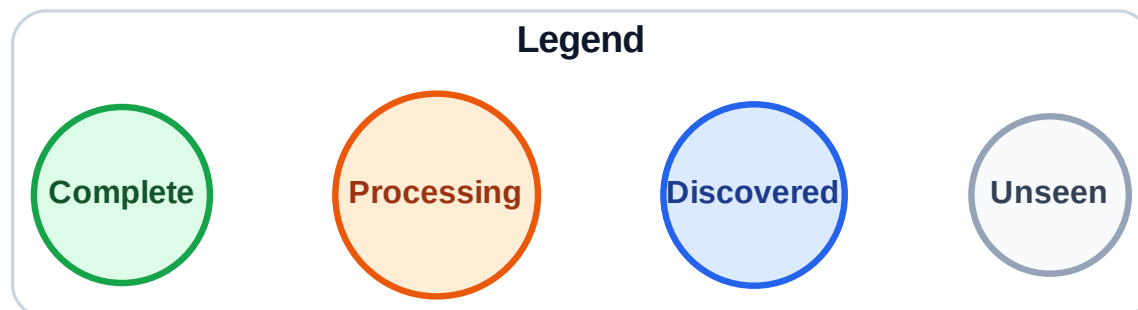
Stack (top at right): J | G | E | D | C | H

Visited: A, B

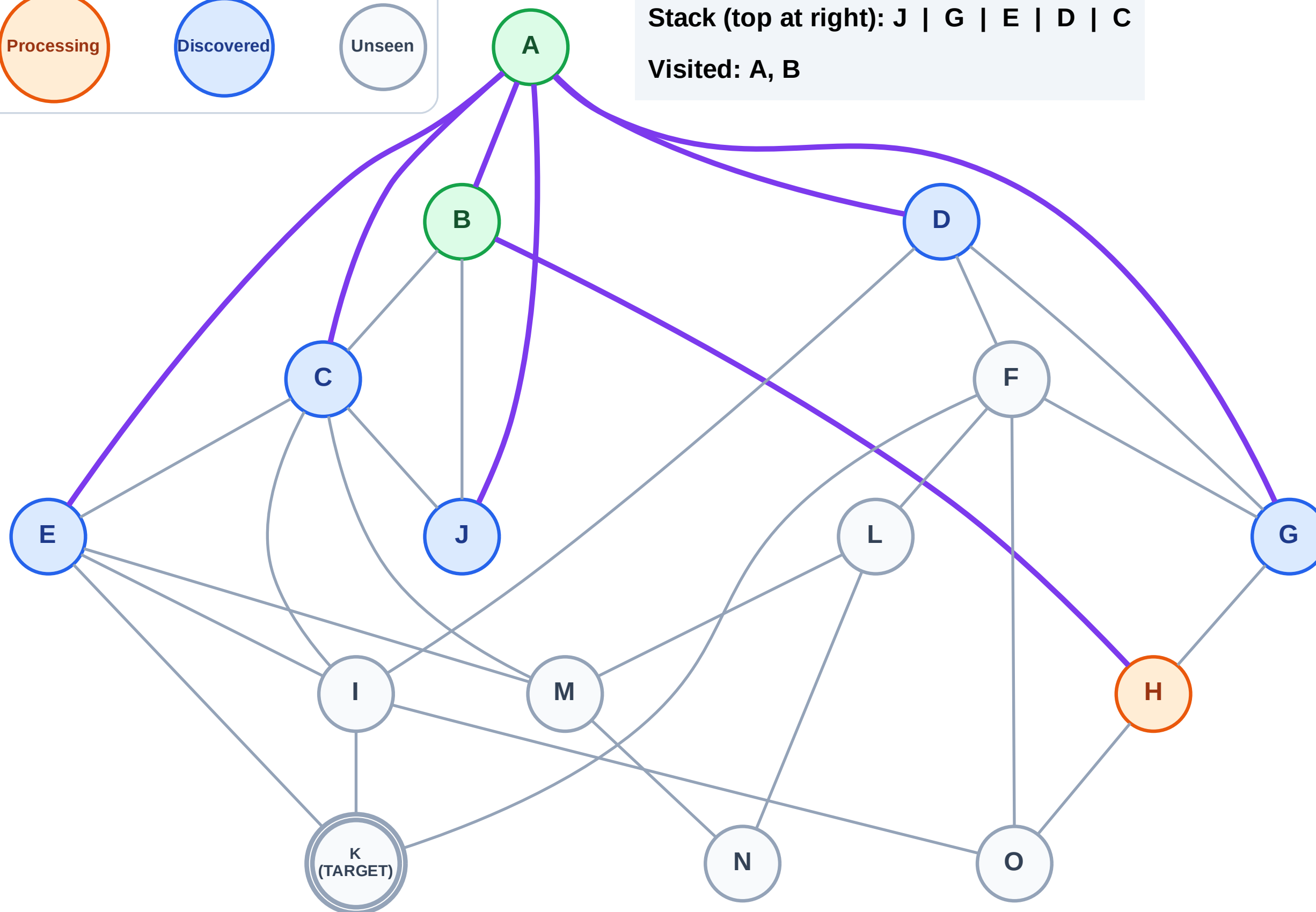


DFS Traversal

Step 6: Process node H



Stack (top at right): J | G | E | D | C
Visited: A, B



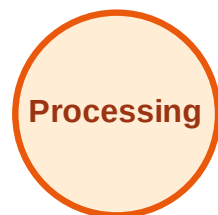
DFS Traversal

Step 7: Discover O from H

Legend



Complete



Processing



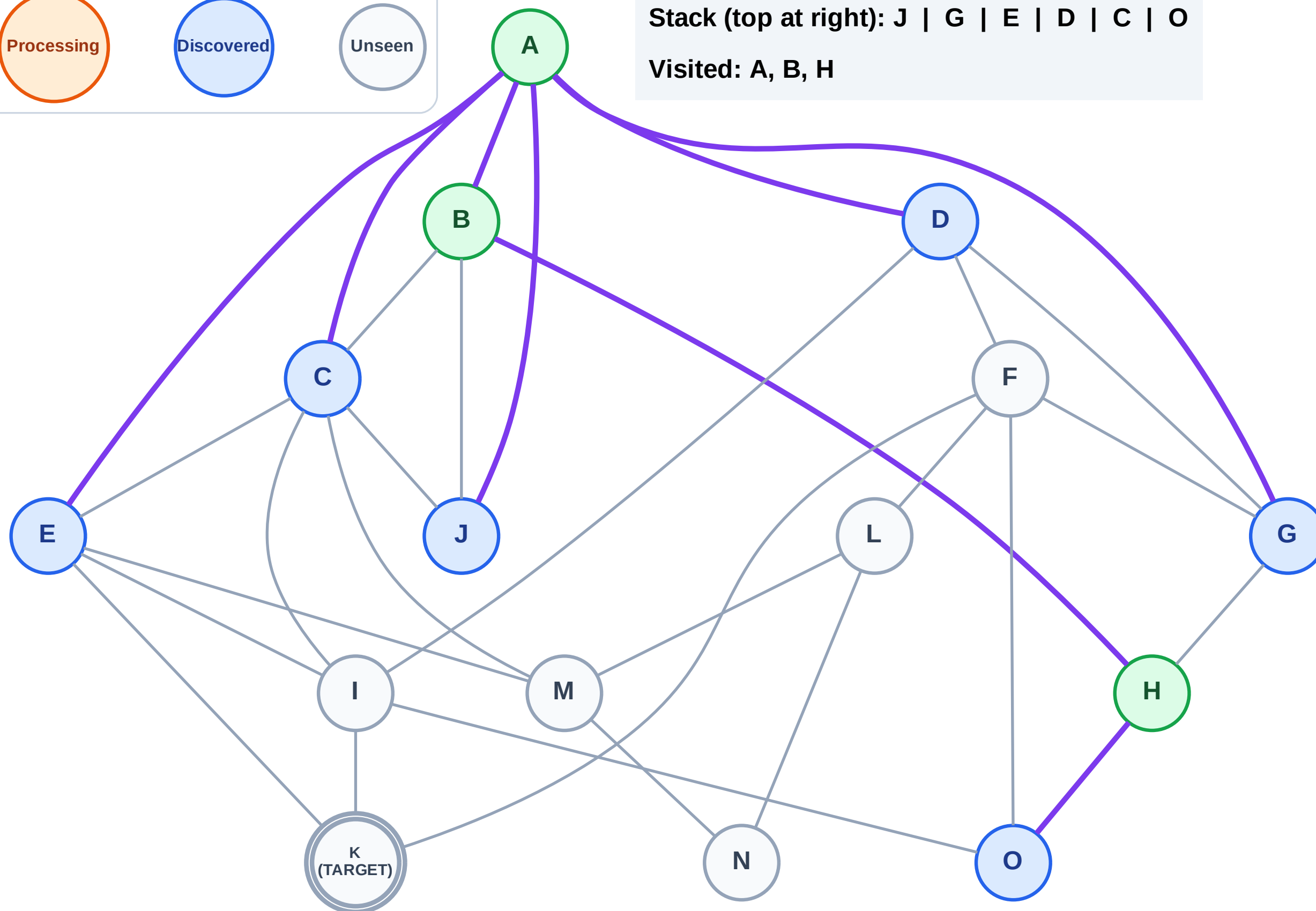
Discovered



Unseen

Stack (top at right): J | G | E | D | C | O

Visited: A, B, H



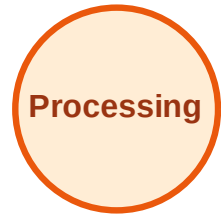
DFS Traversal

Step 8: Process node O

Legend



Complete



Processing



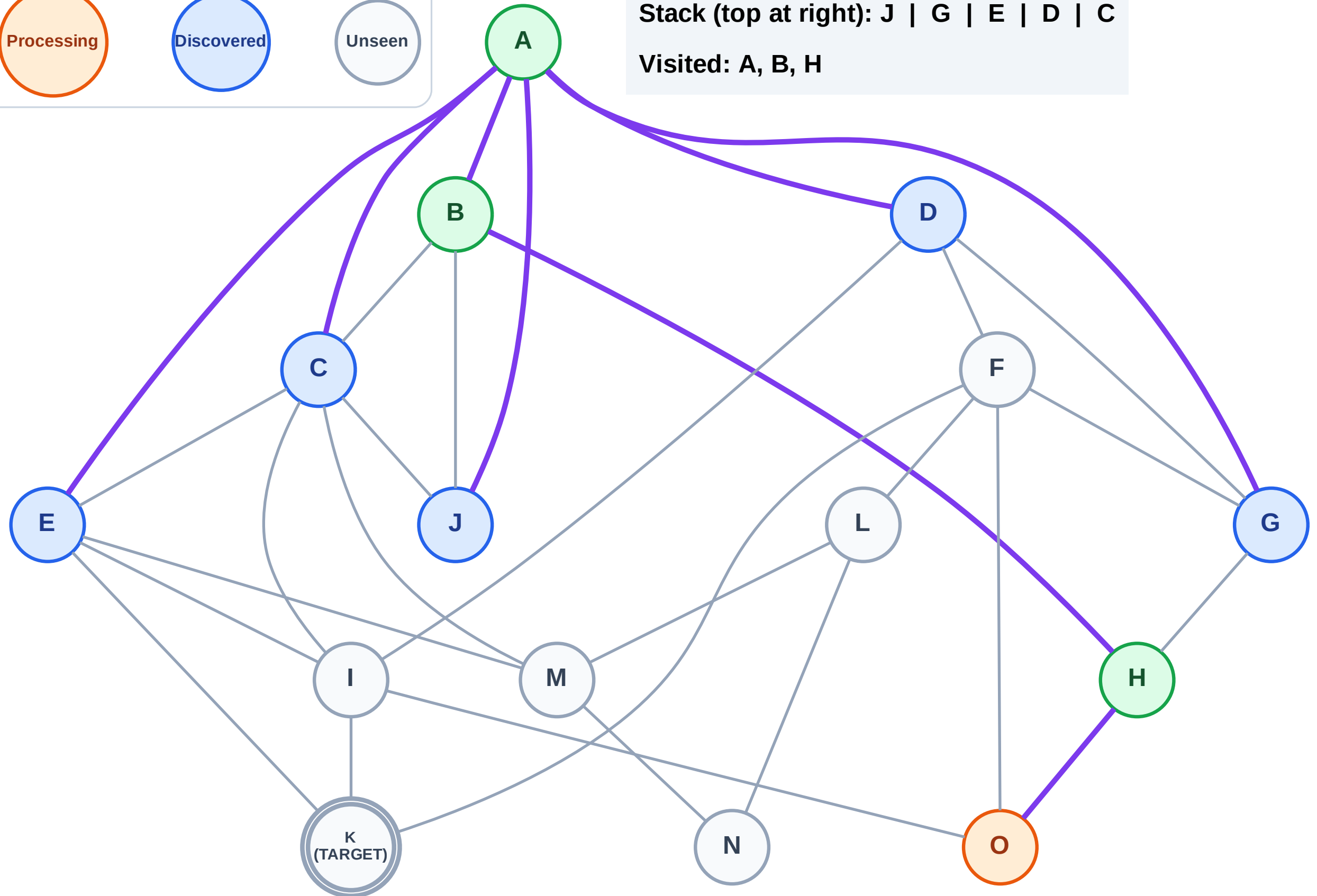
Discovered



Unseen

Stack (top at right): J | G | E | D | C

Visited: A, B, H



DFS Traversal

Step 9: Discover I, F from O

Legend

Complete

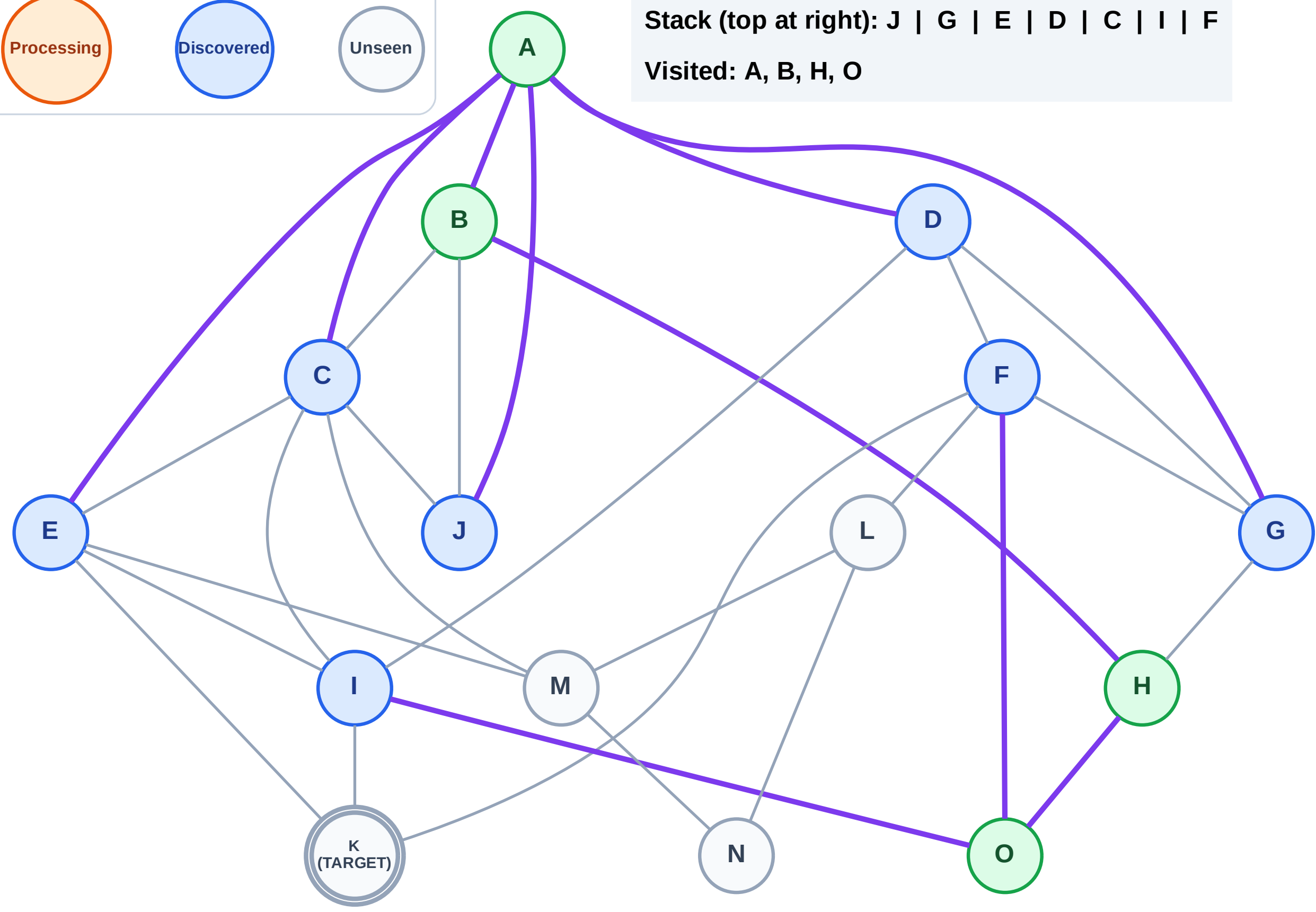
Processing

Discovered

Unseen

Stack (top at right): J | G | E | D | C | I | F

Visited: A, B, H, O



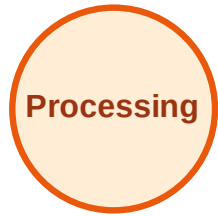
DFS Traversal

Step 10: Process node F

Legend



Complete



Processing



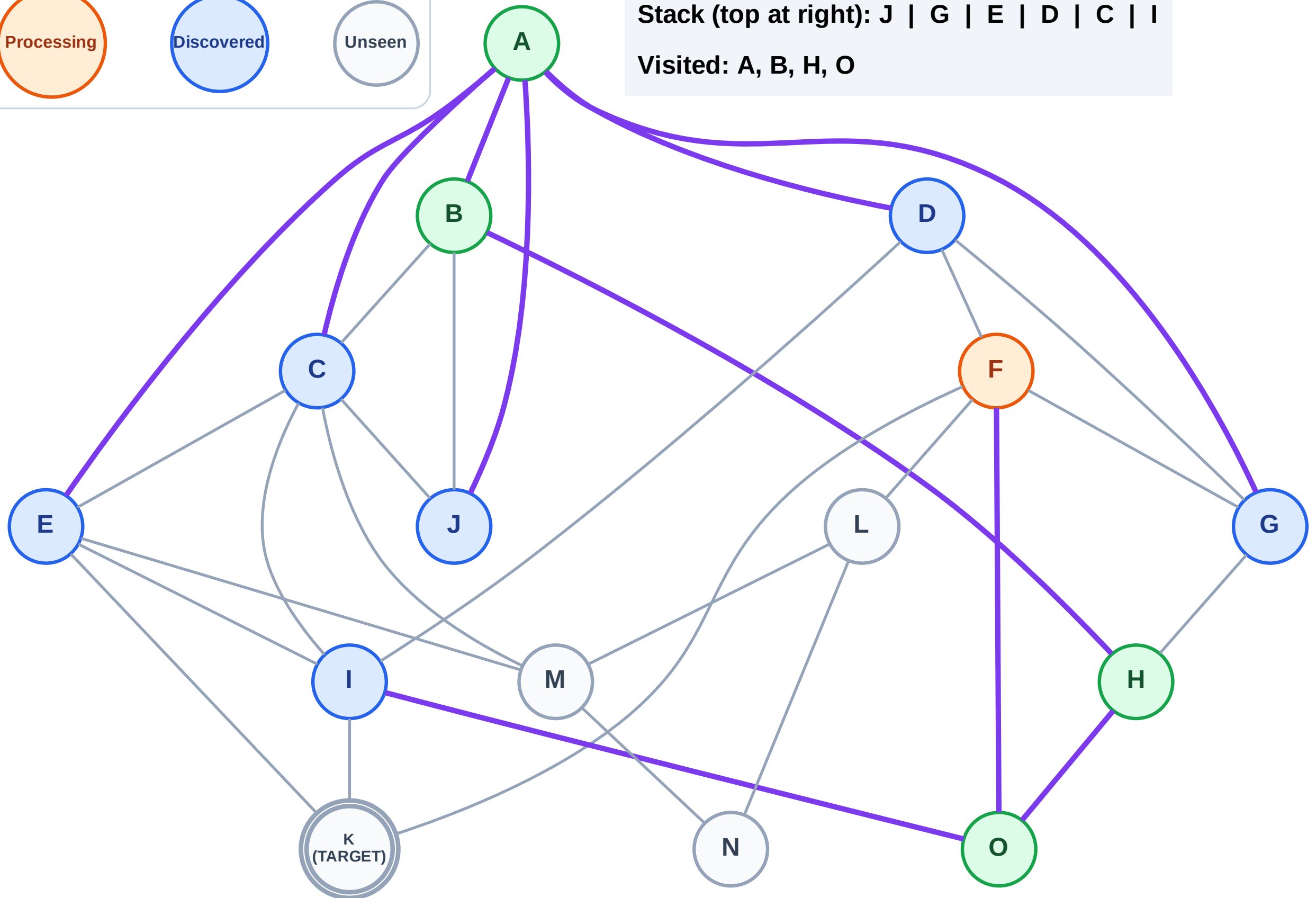
Discovered



Unseen

Stack (top at right): J | G | E | D | C | I

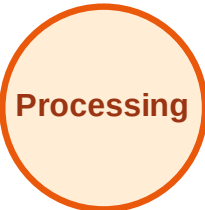
Visited: A, B, H, O



DFS Traversal

Step 11: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (5 hops): A → B → H → O → F → K

Legend



Stack (top at right): J | G | E | D | C | I | L

Visited: A, B, F, H, K, O

